Shishira Bhagavath

Postdoctoral Researcher in Synchrotron Imaging of Advanced Manufacturing Processes

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Professional Experience

Strategic researcher with 10+ years of expertise (PhD+Post PhD) focusing on X-ray imaging, fundamental solidification, image processing and industrial-scale High-Pressure Die Casting (HPDC) and Additive Manufacturing. A proven technical leader with 3 years of experience managing high-impact collaborations between elite academic institutions (Imperial College) and global industry leaders (Ford Motor Company, ESA, Renishaw Plc.). Dedicated to driving sustainability through novel material discovery and the engineering of next-generation, low-carbon manufacturing processes in the automotive and aerospace sectors.

Education

Indian Institute of Technology Bombay — Mumbai, India
PhD in Mechanical Engineering — *Aug. 2017 – Jan. 2022*
Thesis: Formation of porosity during semisolid deformation of Light Alloys

National Institute of Technology Jamshedpur — Jamshedpur, India
M. Tech in Foundry Technology (CPI: 9.69/10, Rank 1/16) — *Aug. 2013 – May 2015*

RV College of Engineering — Bengaluru, India
Bachelor of Engineering in Mechanical Engineering (CPI:8.77/10) — *Aug. 2009 – May 2013*

Experience

Post Doctoral Researcher — Feb. 2022 – Present
MXI Lab, University College London, — *London, UK*

- Leading a team of 8 researchers working on Laser Powder Bed Fusion (LPBF) and supervising their day-to-day activities. Co-managing research activities related to Computed Tomography, LPBF and several Spectroscopy techniques. Research activities include novel material development for LPBF, Directed energy deposition of Copper, Machine learning model to predict porosity in High Pressure Die Casting (HPDC) of automotive components and Synchrotron imaging of various industrial applications.

Research Associate — Sep. 2021 – Jan. 2022
Indian Institute of Technology Bombay, — *Mumbai, India*

- Led a team of three researchers working on the solidification process modeling and experimentation.

Teaching Assistant — Jan. 2018 – Jan. 2022
SAIF, Indian Institute of Technology Bombay, — *Mumbai, India*

- Performed CT scanning and image analysis of various in-house and industrial users.

Assistant Professor — July 2015 – June 2016
GITAM School of Technology Bengaluru, — *Bengaluru, India*

- Developed 'Fundamentals of Numerical Modelling' course. Taught Engineering Drawing (1st year undergraduate), Manufacturing Process (2nd year undergraduate) and Automotive Engineering (3rd year undergraduate).

Projects

Porosity prediction of HPDC Automobile Components — Aug. 2016 – Present

- Co-developed a machine learning model for porosity prediction in High Pressure Die Cast(HPDC) for automotive components for Ford Motor Company, USA.
- Developed a physical twin to mimic HPDC process capable of observing microstructural changes *in situ* during the process using Synchrotron X-ray radiation.
- Proposed a new mechanism of convection-induced gas porosity in pressurised castings.
- Developed a deformable grid 1D model to predict gas porosity sizes in isolated liquid channels.
- Implemented data curation methods to handle data from experiments, computed tomography and CFD models that served as ground truths for the ML model.
- Supervising 2 PhD students, assisting in data analysis and paper writing. Project management and engagement with industrial partner (Ford Motor Company, USA)

Sustainable 3D printing of Copper using Chemically modified Graphene June 2023 – Present

- Developed a new graphene-based additive to improve the LPBF and DED printability of copper. Leading commercialisation efforts to scale up this technology from lab to pilot scale to develop next-gen heat exchangers for Electrical Vehicles and Data Centres.
- Developed optimised printability, defect and surface process maps for LPBF and DED.
- Developed complete powder and print part characterisation protocols for certification.

Materials Made Smarter Centre Feb. 2022 – Feb. 2025

- Led and coordinated nine synchrotron experiments of advanced manufacturing processes,e.g., DED, LPBF and HPDC.
- Involved in developing a new mechanism for porosity formation and growth in DED and proposed two industrial solutions to reduce porosity in DED components.
- Characterised various types of Automated Dry Fibre Placement (ADFP) composites.
- Led a team of six to develop synchrotron-informed process maps of novel aluminium, copper alloys and commercial titanium and nickel superalloys, FeSi soft magnets.
- Developed printing strategy to LPBF novel catalytic material on copper substrate and characterised its functionality.

QUAD-Insitu and Operando Process Replicator Aug. 2024 – Present

- Overseeing maintenance of a custom-built LPBF machine. Performed routine calibration,beam profiling and part replacement/upgrade.
- Co-led upgradation activities to install new sensors and laser scan head.

Solid-state 3D Printing of Aluminium Alloys Feb. 2023 – Feb. 2024

- Developed a novel methodology to estimate the yield of production of friction stir 3D printed Al alloy using CT scans.

Novel Impact Resistant Metal Matrix Composite Metallic Foams Feb. 2018 – Jan. 2022

- Established casting and characterisation facility to fabricate novel *in situ* alloyed metal matrix composites.
- Manufactured Al and Zn composite foams and determined their mechanical properties.

Sustainable use of Integrated Steel Plant Wastes June 2014 – May 2015

- Developed novel Blast furnace slag briquette cements and tested mechanical properties and corrosion resistance.
- Performed pilot scale investigation to study the effectiveness of iron ore waste + low-grade coal mixture pellets.

Select Publications: 21 published|7 under review|Full list in Google Scholar

- **Shishira Bhagavath**, Yinglun Hong, Baoshun Zhou, Yuhan Wang, Kwan Kim, Anna CM Getley, Samy Hocine, Wei Li, Mark Isaacs, Michal Klosowski, Alexander Rack, Sarah Fearn, Chun Huang, Pedram Asef, Mohamad Bayat, Eduardo Saiz, Iuliia S Tirichenko, Peter D Lee, Chu Lun Alex Leung, *"Synchrotron-informed process mapping of Copper and Copper with Chemically Modified Graphene"*, Carbon, 258 (2026), 121835.| Link
- **S.Bhagavath**, B.Cai, R.C.Atwood, M.Li, B.Ghaffari, P.D.Lee, S.Karagadde, *"Combined deformation and solidification driven porosity formation in light alloys"*, Metallurgical and Materials Transactions A, 50 (2019), 4891–4899.| Link
- **S.Bhagavath**, Z.Gong, T.Wigger, S.Shah, B.Ghaffari, M.Li, S.Marathe, P.D.Lee, S.Karagadde, *"Role of the local stress systems on microstructural inhomogeneity during semisolid injection"*, Acta Materialia, 214 (2021), 117015.| Link
- **S.Bhagavath**, Z. Gong, T. Wigger, B. Ghaffari, M. Li, S. Marathe, P.D. Lee, S. Karagadde, *"Mechanisms of gas and shrinkage porosity formation in solidifying shear bands"*, Journal of Materials Processing Technology, 299 (2022), 117338.| Link
- Ujjal Tewary, Dennis Paul, H.K. Mehtani, **Shishira Bhagavath**, Alankar Alankar, Goutam Mohapatra, Satyam S. Sahay, Ajay S. Panwar, Shyamprasad Karagadde, Indradev Samajdar, *"The origin of graphite morphology in cast iron"*, Acta Materialia, 226 (2022), 117660.| Link
- U.Godwal, **S.Bhagavath**, S.Roy, B.Ghaffari, M.Li, L.Godlewski, P.D.Lee, S.Karagadde, *"A knowledge-transfer based multiscale machine learning framework for predicting solidification process defects"*, Journal of Manufacturing Processes, 154 (2025), 82-93.| Link
- Kai Zhang, **Shishira Bhagavath**, Sebastian Marussi, Alexander Rack, Robert Atwood, Martyn A. Jones, Chu Lun Alex Leung, Chinnapat Panwisawas, Junji Shinjo, Peter D. Lee, *"Pore-lean directed energy deposition additive manufacturing through laser power modulation"*, Acta Materialia, 302 (2025), 121515| Link

Grants:3 as Co-Investigator (£652k), Synchrotron experiment grants (£380k)

- **Principal-Investigator:** Research England's Higher Education Innovation Fund (HEIF) project (KEI2026-01-17 BHAGAVATH), *"Performance and Resource Efficient Manufacturing via Intelligent Ultra functional Materials using Directed Energy Deposition (PREMIUM DED) "*, Funded value: £29,951
- **Principal Investigator:** ESRF beamtime grant, MA-7161, £80,000 in machine time.
- **Co-Investigator:** Beamtime grants at Diamond Light Source, UK, £300,000 in machine time.
- **Co-Investigator:** UCL Research Culture Seed Fund, Project title: "Student Training and Experience for Proposals (STEP)", Funded value: £2,000.
- **Co-Investigator:** EPSRC Impact Acceleration Account (IAA) project (KEI2024-01-01 EPSRC IAA D2U LEUNG), *"Enhanced feedstock for additive manufacturing copper components – EFFICIENT"*, Funded value: £135,992.
- **Co-Investigator:** Made Smarter Innovation – Impact Booster, Innovate UK (10154949), *"Enhanced feedstock for additive manufacturing copper components – EFFICIENT"*, Funded value: £513,930.
- **Principal Investigator:** Royce grant, Use of Aconity Mini LPBF machine to manufacture novel copper CMG test cubes. Funded value: £7,000 in machine time.

Awards and Professional Service

- **Editor's Choice:** Metallurgical and Materials Transactions A (MMTA) and shortlisted for the 2020 Henry Marion Howe Medal for the article: "Combined deformation and solidification driven porosity formation in light alloys" (2019).
- **Ph.D. Annual Progress Seminar Award:** Department of Mechanical Engineering, IIT Bombay (2020).
- **Best Poster Award:** National Metallurgist's Day (NMD) Annual Technical Meet, Goa, India (2017).
- **Gold Medal for Academic Excellence:** Postgraduate course, 6th Convocation of NIT Jamshedpur (2016).
- **Invited Speaker:** AICTE-QIP-PG Certificate Programme on "Advanced Metallic Alloys for Energy Storage Applications", SVNIT Surat, India, October 2025.
- **Invited Speaker:** *Synchrotron imaging of manufacturing processes-Insights and mechanisms*, IIT Bombay, India, September 2024.
- **Invited Speaker:** *Experimental and numerical investigations of porosity defect in high-pressure die casting*, Ford Motor Company, Dearborn, USA, March 2023.
- **Invited Speaker:** *X-ray imaging of die-shape effects on defect formation during pressurised casting.* Defects and Properties of Cast Metals Symposium, TMS, Orlando USA, March 2023.
- **Invited Speaker:** Application of Soft-computing techniques and Numerical Modelling in "Additive Manufacturing and Materials Processing" (A2MP 2022), SVNIT Surat, India, December 2022.

Technical Skills

Programming Languages: Python, MATLAB, C, ImageJ Macros, SQL
Characterisation: Optical microscopy, SEM-EDS, XPS, UV-Vis Spectroscopy, Raman spectroscopy, DSC, PIV, CT, Synchrotron Imaging, XRD.
Analysis/Design Tools: OpenFoam, Flow3D, Avizo, Ansys, SolidWorks, Catia, CasaXPS.
Enterprise Training: SPERO: Introduction to thinking entrepreneurially about your career, SPERO 1: Develop your entrepreneurial knowledge, skills and mindset, I/O lab, Test, Validate, Pitch by Aspire Team/UCLB.